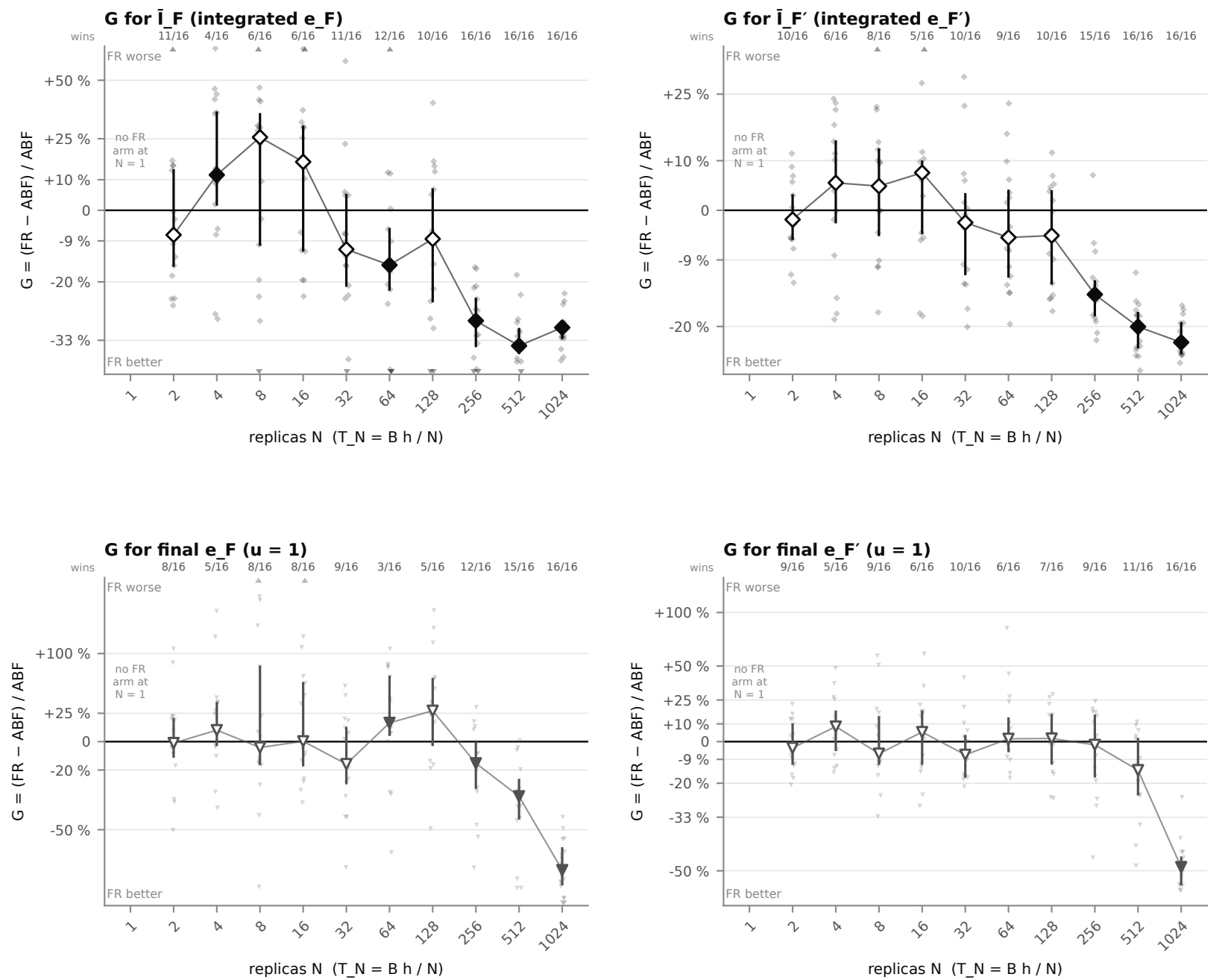


ethane/LTA 150 K: FR relative improvement G(N) at equal budget

Equal total budget per arm per seed: $B = N \times n_steps = 3.072e+08$ walker-steps at every N ($h = 0.0002$, $T_N = B \cdot h / N$); 16 seeds per (N , method). Paired per seed (same N and seed: shared initial conditions and Langevin noise).



ethane/LTA 150 K · S3_fr_gain_vs_N · status: all planned (N, seed, method) complete · FR is not defined at $N = 1$ (ABF only, by design) · LTA $N < 50$: FR uses the finite- N cap extension $\max(1, \text{floor}(0.02 N))$.
Marker: median of the per-seed G ; bar: seed-bootstrap 95% CI of the median (10000 resamples); filled = CI excludes 0; faint: per-seed G (caret: off scale). Log-ratio axis (FR/ABF) labelled in % change. 'wins' row: seeds with $FR < ABF$ / paired seeds. Red 'FR inactive' / 'no death k/n': FR arms that realised no death (equal to ABF), where $G = 0$ is FR inactivity, not equivalence.
Source results/equal_budget_v2/lta_T150/analysis/summary.json (generated 2026-10-10T07:14:18Z, eqb_metrics/2, 10000 bootstrap resamples); plot_synthesis/2.